

Patient-Specific Computational Fluid Dynamics of the Airways

Segmentation, mesh generation, and flow analysis in tracheomalacia

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Computational Biofluid Mechanics

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`github.com/henrivanovermeire/tracheomalacia_cfd`

Clinical and computational question

- Tracheomalacia and congenital stenosis can strongly increase airway resistance.
- CT describes anatomy; CFD estimates velocity, pressure loss, resistance, and flow division.
- Preoperative and postoperative anatomy were reconstructed at corresponding tracheal locations.

Main question

How does the intervention alter lumen calibre and the geometry-driven airflow through the reconstructed airway?

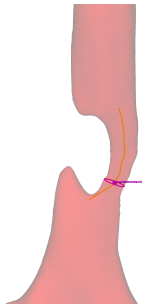


Image to result

DICOM → 3D Slicer/SlicerVMTK → capped STL → Gmsh → OpenFOAM → ParaView

Geometry preparation

- Case-specific airway segmentation
- Centerline-normal measurements
- Centerline-directed extensions and planar caps

github.com/henrivanovermeire/tracheomalacia_cfd [slicer.org](https://www.slicer.org) gmsh.info openfoam.com

Flow solution

- Tetrahedral HXT meshes
- Steady laminar simpleFoam
- Transient postoperative pimpleFoam PoC

Scripted approach for optimal reproducibility

- Repeated processing steps were implemented as version-controlled scripts instead of undocumented manual actions.
- Case generation fixes patch names, units, fluid properties, boundary conditions, numerical schemes, and decomposition settings.
- Post-processing scripts extract residuals, conservative patch flows, pressure loss, resistance, mesh sensitivity, and transient convergence.
- Figure generators use fixed views, scales, sample density, and reporting locations.

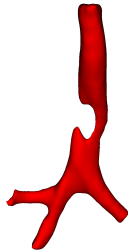
Scripts and documentation: github.com/henrivanovermeire/tracheomalacia_cfd

Traceable chain

Source DICOM and settings ↓
Segmentation and mesh scripts ↓
OpenFOAM case dictionaries ↓
Machine-readable CSV/JSON ↓
Report figures and PDFs

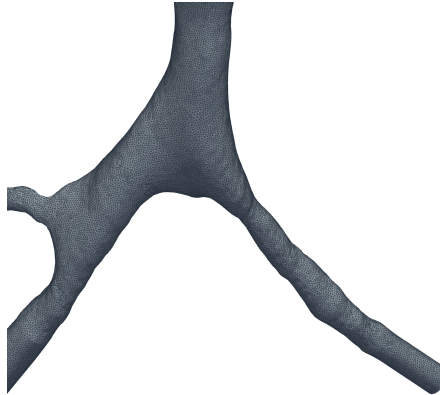
3D Slicer: CT to airway geometry

- Open-source platform for medical-image visualisation, segmentation, and quantitative analysis.
 - CT voxels were selected using connected intensity thresholding:
 - preoperative: -1500 to -300 HU;
 - postoperative: -1500 to -650 HU.
 - Incorrectly included lung air was removed manually.
 - SlicerVMTK centerlines supported centerline-normal measurements and anatomically directed flow extensions.
 - Planar caps produced closed, CFD-ready STL surfaces.
- Geometry is patient-specific, but segmentation and unknown CT respiratory phase remain sources of uncertainty.



Gmsh: surface to finite-volume mesh

- Open-source mesh generator used to convert each closed airway STL into a three-dimensional tetrahedral volume mesh.
- Named physical groups preserve the CFD boundaries: `inlet`, `outlet_1-3`, and `wall`.
- HXT meshes were generated at target sizes of 0.25, 0.20, 0.15, and 0.12 mm.
- Meshes were imported with `gmshToFoam`, scaled from millimetres to metres, and verified using `checkMesh`.



Selected steady mesh

0.15 mm HXT: 776,568 tetrahedra; best observed accuracy–cost compromise.

OpenFOAM: the CFD toolbox

- Open-source finite-volume toolbox for solving conservation equations on general polyhedral meshes.
- Pressure and velocity are solved as coupled physics using segregated algebraic equations.
- Incompressible pressure is kinematic: $p = P/\rho$; reported pressure differences use $\Delta P = \rho \Delta p$.
- Cases were decomposed with Scotch and solved in parallel using MPI.

The same directory structure supported both steady and transient airway simulations.

Case organisation

0/: U , p , initial and boundary conditions

constant/: polyMesh, viscosity, flow model

system/: controlDict, fvSchemes, fvSolution, decomposition

Remote computation and DevOps workflow

- Local workstation: segmentation, case preparation, post-processing, reporting, and Git version control.
- Remote high-performance compute VM: CPU-intensive meshing and OpenFOAM execution.
- Scripted SSH/rsync transfers moved cases to the VM and retrieved results.
- Docker supplied OpenFOAM 2512; MPI distributed decomposed domains and `reconstructPar` reassembled the fields.

Execution pipeline

Git-tracked case → SSH transfer
→ Docker → `decomposePar`
→ MPI solver → `reconstructPar`
→ result retrieval

Provenance limitation

OpenFOAM 2512 and the case files are retained, but the historical Docker digest was not archived; the mutable `latest` tag prevents bit-for-bit environment reproduction.

Solvers used: simpleFoam and pimpleFoam

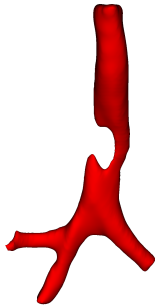
	simpleFoam	pimpleFoam
Purpose	Steady incompressible flow	Transient incompressible flow
Coupling	SIMPLE pressure–velocity iterations	PIMPLE: SIMPLE-like outer loops with PISO-like pressure correctors
Time	Iterations are not physical time	Advances through physical timesteps
Stability	Under-relaxation between iterations	Adaptive Δt with a Courant limit
This study	Final steady result on 0.15 mm HXT mesh	One postoperative cycle on 0.25 mm HXT mesh
Outcome	Converged at iteration 1611	99.826% of timesteps met the outer criterion

Interpretation

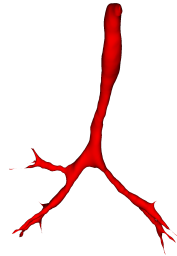
Algebraic convergence controls equation-solution error; it does not by itself establish mesh or timestep independence.

Patient-specific airway reconstruction

Preoperative



Postoperative



The preoperative stenosis was preserved during segmentation and manually separated from included lung air.

Normalized centerline-based pre/post comparison



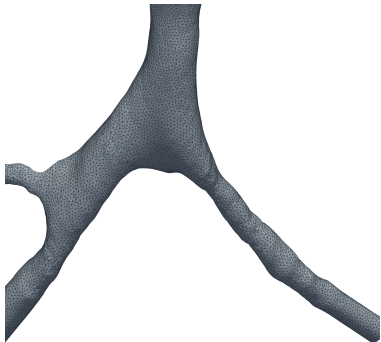
Quantity	Preoperative	Postoperative
Area (mm ²)	1.813	5.169
Minimum Feret diameter (mm)	0.798	2.292
Equivalent diameter (mm)	1.519	2.565

Preoperative deficit relative to postoperative section: 64.9% by area; 65.2% by minimum Feret diameter.

This is a cross-scan comparison, not conventional within-scan stenosis severity.

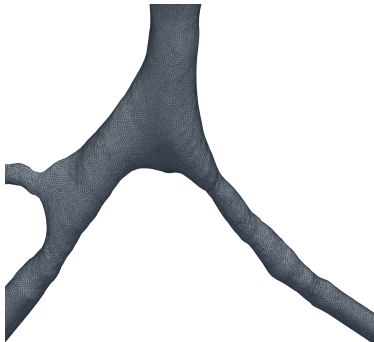
Mesh generation and verification

0.25 mm baseline



180,163 cells

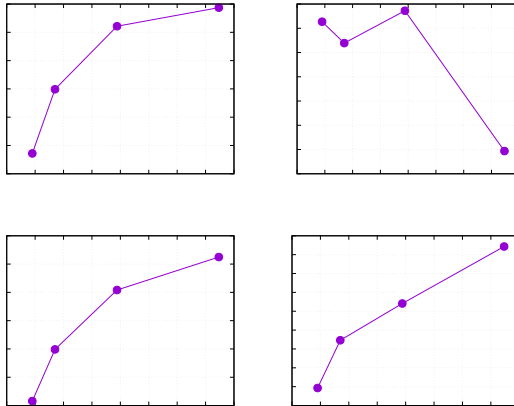
0.15 mm selected



776,568 cells

The 0.15 mm HXT mesh had no faces above 70° non-orthogonality and provided the best accuracy–cost balance.

Mesh sensitivity



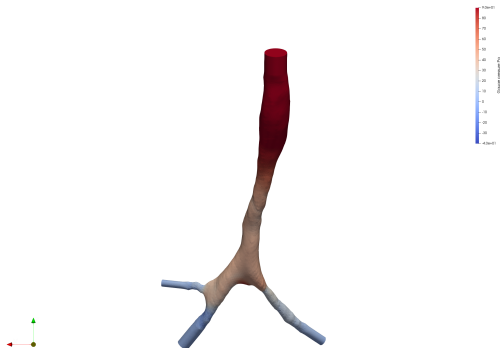
From 0.15 to the unconverged 0.12 mm comparison, monitored values changed by at most 1.83%, while cell count increased by 92% and solver execution time by 157%. This suggests an apparent sensitivity plateau, not proven mesh independence.

Steady postoperative airflow

Velocity



Dimensional pressure



Selected 0.15 mm HXT mesh at 2 L/min: $\Delta P = 35.21$ Pa, conservative-flux resistance = 17.61 Pa/(L min), peak velocity = 9.67 m/s.

Flow distribution emerges from geometry

Computed distribution

$$Q_R = 73.69\%$$

$$Q_L = 26.31\%$$

- Total inlet flow was prescribed.
- Every distal outlet had the same zero gauge pressure.
- No outlet split was imposed:

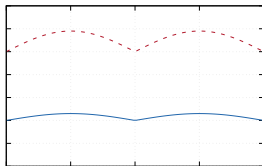
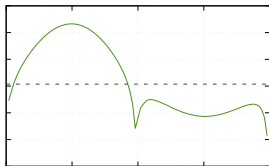
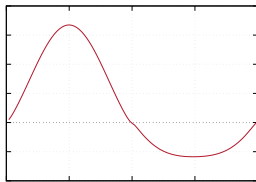
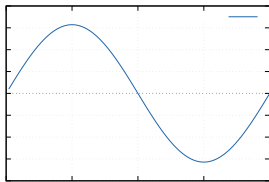
$$Q_i \sim \frac{\Delta P}{R_i}.$$

- Diameter, length, curvature, branch angle, and separation determine effective resistance.

Interpretation limit

This is geometry-driven flow division in the truncated domain, not measured physiological ventilation; peripheral resistance and compliance are absent.

Transient postoperative proof of concept



- Complete 2 s cycle
- Peak flow: ± 6.29 L/min
- ΔP : -58.7 to 167.4 Pa
- R : 5.71 – 26.63 Pa/(L/min)
- Near-zero-flow resistance masked
- 99.826% of timesteps met the PIMPLE criterion

Coarse 0.25 mm PoC, bounded upwind and $Co_{\max} = 5$: qualitative phase trends, not a timestep-independent prediction.

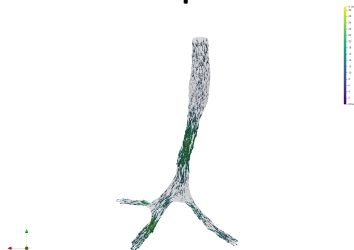
Flow reverses over the breathing cycle

Accelerating inspiration



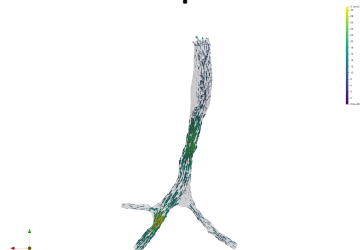
$t = 0.24$ s

Peak inspiration



$t = 0.50$ s

Peak expiration



$t = 1.50$ s

Common 0 m/s to 31 m/s colour scale and vector scale.

Principal findings

- The preoperative area was 64.9% below the normalized centerline-matched postoperative section; respiratory phase was not recorded.
- The selected-mesh model produced a geometry-driven 73.69/26.31% right/left split.
- The 0.15 mm mesh balanced sensitivity and computational cost for steady CFD.
- The transient workflow resolves the imposed breathing phases as a PoC.

Main limitations

- Static, rigid CT geometry
- No peripheral lung impedance
- Laminar model
- No prism boundary layers
- One patient and one initial cycle
- Coarse transient fallback at high Courant number

Questions?